An open pseudo-3D Sleipner benchmark and operating protocol for time-lapse CO_2 support mapping

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Abstract

Public field evaluation of time-lapse seismic change mapping remains limited, especially for multi-inline assessment on open data. We construct a public pseudo-3D Sleipner benchmark from eleven neighboring inlines with two tracks: a p07 temporal branch (1994 to 2001, 2004, 2006) and a p10 direct branch (1994 to 2010). We compare constrained classical baselines, plain and hybrid machine learning, and deterministic structured support reconstruction. On a harder three-seed synthetic benchmark, plain machine learning is more stable than hybrid machine learning (Dice 0.544 ± 0.008 test; 0.206 ± 0.032 out-of-distribution). On field data, structured support reconstruction improves the main p10 support-volume IoU from 0.562 to 0.685 while preserving trace-support IoU at 0.907. Reusing three saved seed artifacts on the field benchmark keeps the same p10 ordering (support-volume IoU 0.637 ± 0.037 for structured plain machine learning versus 0.526 ± 0.048 for plain machine learning) and shows that on p07 the structured variant improves support-volume IoU over plain machine learning (0.440 ± 0.052 versus 0.304 ± 0.061) while constrained classical retains the highest raw p07 support-volume IoU. We then add a fixed-artifact operating-point protocol for wave-temporal deployment: in a three-seed quantile sweep over 0.74 to 0.80, only 0.74 and 0.75 satisfy a joint gate (p10 support-volume IoU above constrained classical and p07 support-volume IoU at least as high as constrained classical), with 0.74 selected as primary. Adaptation sanity checks show that low-memory field adaptation can collapse occupancy (IoU to

0, outside-support fraction to 1), motivating no-adaptation transfer for this setting. The benchmark and protocol provide a reproducible open-data basis for field-facing monitoring claims.

Keywords. CO_2 storage monitoring; time-lapse seismic; Sleipner; pseudo-3D benchmark; machine learning; structured support mapping

1 Introduction

Time-lapse seismic data have played a central role in monitoring the Sleipner CO_2 storage site for nearly three decades [3]. The site is important not only because of its long injection history, but also because it has become one of the rare public field datasets where repeated seismic vintages, benchmark reservoir information, and interpreted plume products are available together [8, 9]. This makes Sleipner a natural test case for methods that aim to move beyond qualitative difference maps toward more quantitative and reproducible change mapping.

At the same time, Sleipner is a difficult testbed. Repeated surveys are affected by non-repeatability, processing changes, tuning effects, and uncertainty in how seismic response relates to plume geometry [1, 6, 7]. Recent work has shown that physics-based inversion and segmentation constraints can improve field interpretation, but the workflow remains demanding and the final field products still depend on strong structural information [2, 7]. Machine learning studies have also shown promise, including prior work on Sleipner interpretation and site-specific field transfer at other storage sites [4, 5]. However, there is still a gap between simplified synthetic demonstrations and open, field-facing, reproducible pseudo-3D evaluation on public CO_2 monitoring data.

This paper addresses that gap with three linked contributions. First, it constructs an open pseudo-3D Sleipner benchmark from eleven neighboring inlines while keeping the 2007 and 2010 processing families separate, thereby creating a transparent multi-inline testbed for public field evaluation. Second, it defines benchmark-constrained structured support mapping from per-section predictions to pseudo-3D support volumes, so that section-level outputs can be assessed as coherent monitoring products rather than as isolated slices. Third, it introduces an operating-point selection protocol for cross-benchmark field deployment using three-seed quantile sweeps, explicit

gate criteria against constrained classical baselines, and adaptation sanity checks.

The central research question is whether benchmark-constrained pseudo-3D support mapping on public Sleipner data can improve on plain per-section machine learning, provide a defensible comparison with constrained classical baselines, and be made reproducible through an explicit operating protocol. On the present benchmark, the clearest field gain is obtained with structured support reconstruction built on plain machine learning, especially on the direct p10 branch, while the new operating protocol defines where wave-temporal deployment is reliable across both p10 and p07. The synthetic and field evaluations place the hybrid model in a secondary role, so the contribution should be read primarily as a benchmark, support-mapping, and deployment-protocol study.

2 Public data and benchmark construction

2.1 Public Sleipner data

All field experiments use public data from the Sleipner 4D Seismic Dataset and the Sleipner 2019 Benchmark Model distributed through CO2DataShare [8, 9]. The seismic dataset contains processed 4D seismic cubes from 1994 to 2010 and includes different processing families such as the 2007 and 2010 products. The benchmark model provides the reference simulation grid, velocity trends, reservoir horizons, and plume boundary products that allow public structural constraints to be aligned with the seismic data.

The benchmark construction uses inline exports from the processed seismic cubes. This choice is intentional and is aimed at establishing a transparent pseudo-3D monitoring benchmark from open data.

2.2 Pseudo-3D benchmark design

We construct two related but distinct field benchmarks from eleven contiguous inlines centered on inline 1840. The temporal benchmark uses the 2007 processing family and evaluates the 1994 baseline against the 2001, 2004, and 2006 monitor vintages. The direct benchmark uses matched 2010 processing and evaluates the 1994 baseline against the 2010 monitor. These two benchmarks are kept separate throughout the study because mixing processing

families would confound temporal interpretation.

For every exported inline pair, we store a normalized baseline section, a normalized monitor section, a storage-interval mask derived from the public benchmark surfaces, a trace-support product based on the public 2010 plume outlines, and a 2D support-volume proxy aligned to the seismic section. The result is a pseudo-3D volume with dimensions inline, vintage, sample, and trace. This arrangement allows crossline continuity to be evaluated explicitly rather than inferred from a single section.

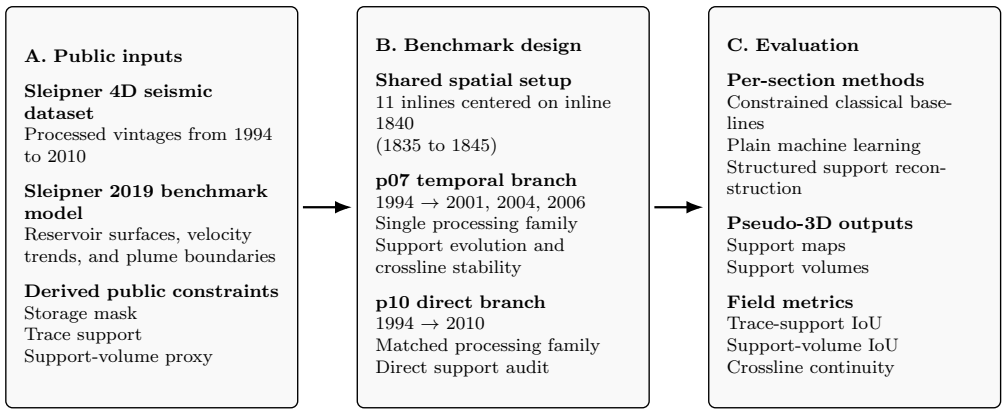

Figure 1: Public pseudo-3D benchmark design and evaluation protocol. Panel A summarizes the public seismic and benchmark products used to derive structural constraints. Panel B defines the two benchmark tracks used in the study. Panel C shows how constrained classical baselines, plain machine learning, and structured support reconstruction are evaluated through pseudo-3D support outputs and field metrics.

2.3 Harder synthetic benchmark

The synthetic benchmark used here serves as a stress test for model stability. Compared with the earlier toy benchmark, the current synthetic benchmark introduces stronger mismatch behavior, negative controls, and out-of-zone changes. In particular, we evaluate clean examples, harder out-of-distribution examples, and negative-control scenarios including no-change, mismatch-only, and out-of-zone cases. These synthetic experiments make it possible to evaluate model stability and false alarms across multiple random seeds in a way that cannot be achieved on the fixed field data alone.

3 Methods

3.1 Method overview

The study design is summarized in Fig. 1. Public seismic and benchmark products are first converted into a pseudo-3D benchmark, after which constrained classical baselines and machine learning models are evaluated under the same structural constraints. The field-facing output is a support-mapping product defined on a shared pseudo-3D benchmark.

3.2 Per-section methods

We compare three method families at the per-section stage. The first consists of deterministic classical baselines such as difference maps, cross-equalized differences, local warping, and a matched warp-shift comparator. These methods provide a practical baseline under the same benchmark constraints used for machine learning outputs. The second family is plain machine learning, which predicts change support from baseline and monitor sections without additional hybrid inputs. The third family is a hybrid model that was originally expected to improve robustness.

The current evidence places the hybrid model in a comparator role rather than in the central field result.

3.3 Structured support reconstruction

The strongest field method evaluated here is the structured plain variant. It begins with per-section probabilities from the plain machine learning model, applies the public storage-interval mask, and then performs deterministic support reconstruction to promote laterally coherent and geologically plausible support bodies. The reconstruction stage removes isolated artifacts, preserves thin but coherent support streaks, and performs connected-component cleanup in pseudo-3D space across inline, sample, and trace dimensions.

This stage is intentionally simple. It transforms per-section probabilities into a support volume that is better aligned with the public benchmark constraints and more useful for multi-inline monitoring assessment.

We also tested a layer-aware extension that split the storage interval into benchmark-informed bands and enforced stronger vertical rules. That

extension did not improve the real benchmark and is therefore treated as a secondary result.

3.4 Evaluation metrics

Synthetic experiments are reported using Dice, intersection-over-union, false-positive rate, and expected calibration error. Because the synthetic benchmark includes repeated training, the machine learning models are summarized with three-seed mean and standard deviation. Deterministic classical baselines are reported once because they do not depend on training seeds.

Field experiments are evaluated with metrics that are more appropriate for pseudo-3D monitoring. These include trace-support intersection-over-union with the public 2010 support proxy, support-volume intersection-over-union, crossline continuity, predicted support fraction, and predicted fraction outside the support volume. The direct 2010 benchmark is treated as the main truth test because it compares seismic data and support proxy within the same processing family. The p07 temporal benchmark is used as a secondary test of support evolution and crossline stability.

4 Experimental design

The experiments are organized around four questions. The first asks whether the per-section models are stable on the harder synthetic benchmark. The second asks whether structured support reconstruction improves pseudo-3D field mapping on the direct eleven-inline benchmark. The third asks whether the same field method yields physically sensible temporal behavior on the eleven-inline p07 benchmark. The fourth asks whether a fixed-artifact quantile protocol can identify cross-benchmark operating points that outperform constrained classical baselines while avoiding unstable adaptation behavior.

For synthetic evaluation, we retrain the plain and hybrid models with seeds 11, 12, and 13. For field evaluation, all methods operate on the same exported pseudo-3D benchmarks and the same public support constraints. To test whether the main field claim is tied to a single trained artifact, the p07 and p10 benchmarks are also re-evaluated using the saved seed-11, seed-12, and seed-13 artifact roots from the harder synthetic sweep, with no additional field training. This keeps the comparison focused on the mapping method rather than on manual changes to the data pipeline.

Table 1: Synthetic benchmark results. Classical results are deterministic. Machine learning results are reported as mean ± standard deviation across three seeds.

Split	Method	Dice	IoU	False-positive rate	ECE
Test	Difference	0.099	0.052	0.140	0.143
Test	Plain ML	0.544 ± 0.008	0.374 ± 0.008	0.011 ± 0.003	0.048 ± 0.016
Test	Hybrid ML	0.494 ± 0.064	0.331 ± 0.059	0.012 ± 0.003	0.043 ± 0.013
OOD	Difference	0.108	0.057	0.065	0.073
OOD	Plain ML	0.206 ± 0.032	0.115 ± 0.020	0.004 ± 0.001	0.034 ± 0.019
OOD	Hybrid ML	0.166 ± 0.024	0.091 ± 0.014	0.006 ± 0.002	0.030 ± 0.013

5 Results

5.1 Three-seed synthetic evaluation

The three-seed sweep shows that plain machine learning is currently the more stable and stronger learner on the harder synthetic benchmark. On the test split, plain machine learning reaches a mean Dice of 0.544 ± 0.008, compared with 0.494 ± 0.064 for the hybrid model. On the out-of-distribution split, plain machine learning reaches 0.206 ± 0.032, compared with 0.166 ± 0.024 for the hybrid model. The strongest deterministic classical baseline on the out-of-distribution split remains the simple difference map at 0.108 Dice. These results are summarized in Table 1.

The synthetic sweep clarifies the model comparison and identifies plain machine learning as the base learner that should be carried into the field evaluation.

5.2 Direct eleven-inline benchmark

The direct p10 benchmark is the clearest field result in the paper. Table 2 shows that the structured support reconstruction improves support-volume intersection-over-union from 0.562 for constrained plain machine learning to 0.685 while keeping trace-support overlap nearly unchanged at 0.907 versus 0.910. The constrained classical baseline is far less selective. Its trace-support intersection-over-union is only 0.322 and 62.3% of its predicted support lies outside the public support volume.

Figure 2 illustrates this result on inline 1840. The structured method

Table 2: Direct eleven-inline p10 benchmark. The structured support method yields the strongest support-volume result while preserving lateral support alignment.

Method	Trace-support IoU	Support-volume IoU	Crossline continuity	Fraction outside support
Best classical constrained	0.322	0.363	0.987	0.623
Plain ML constrained	0.910	0.562	0.973	0.018
Plain ML structured constrained	0.907	0.685	0.977	0.025
Plain ML layered structured constrained	0.758	0.475	0.789	0.070

is more compact and better aligned with the benchmark support than the classical baseline, while the layer-aware extension visibly under-recovers the interpreted support extent.

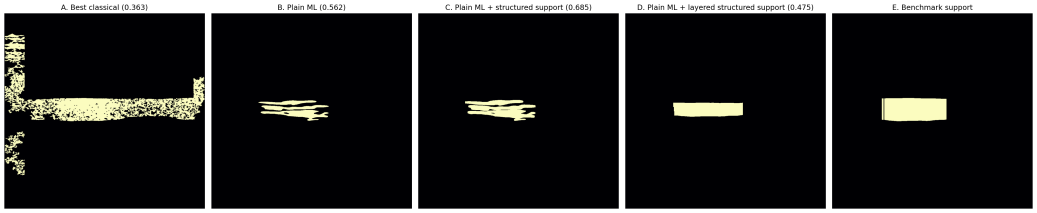

Figure 2: Direct p10 benchmark on inline 1840. From left to right the panels show the best constrained classical baseline, constrained plain machine learning, constrained plain machine learning with structured support reconstruction, the tested layer-aware extension, and the benchmark support. The structured support procedure provides the strongest support-volume agreement on the direct benchmark.

5.3 Temporal eleven-inline benchmark

The p07 temporal benchmark is more nuanced and is therefore important for an honest interpretation. Across the full eleven-inline volume, the constrained classical baseline achieves a higher raw support-volume intersection-over-union than plain machine learning because it predicts support almost everywhere inside the permitted region. Its predicted trace fraction is 0.995 and 48.8% of the predicted support still lies outside the public support proxy. In contrast, structured plain machine learning reaches support-volume intersection-over-union of 0.438 with trace-support overlap of 0.879 and crossline continuity of 0.982, while keeping the predicted fraction outside support to 0.002. Table 3 summarizes these overall results.

Table 3: Overall results on the eleven-inline p07 temporal benchmark. The classical baseline attains a large support-volume score by spreading support too broadly, whereas the structured plain method remains selective and coherent.

Method	Trace-support IoU	Support-volume IoU	Crossline continuity	Fraction outside support
Best classical constrained	0.385	0.477	0.998	0.488
Plain ML constrained	0.873	0.300	0.967	0.001
Plain ML structured constrained	0.879	0.438	0.982	0.002
Plain ML layered structured constrained	0.721	0.433	0.788	0.004

The temporal slices in Fig. 3 show that the structured method preserves a compact and coherent support geometry across the 2001, 2004, and 2006 monitors while following the public 2010 support envelope. The result remains benchmark-constrained and is interpreted here as a support-mapping product.

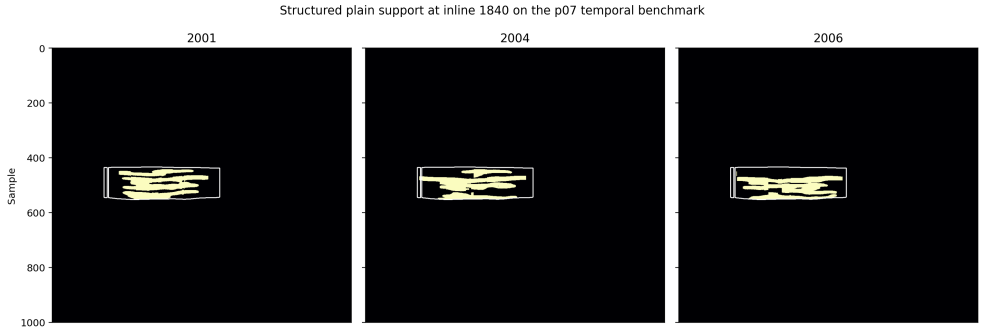

Figure 3: Structured plain support maps on inline 1840 for the p07 temporal benchmark. White contours show the public 2010 support proxy. The maps remain compact across vintages while staying inside the benchmark-constrained support region.

5.4 Three-seed field stability of the main field baselines

To test whether the main field conclusion depends on a single trained artifact, we re-evaluated the direct p10 and temporal p07 benchmarks with the saved seed-11, seed-12, and seed-13 artifacts from the harder synthetic sweep and no additional field training. Table 4 shows that the structured plain method remains the strongest support-volume baseline on p10, improving mean support-volume IoU from 0.526 ± 0.048 for plain machine learning

Table 4: Three-seed field stability check using the saved seed-11, seed-12, and seed-13 artifacts and no additional field training. Values are mean ± standard deviation across the three seed artifacts.

Benchmark	Method	SV IoU	Trace-support IoU	Crossline continuity	Fraction outside support
p07	Best classical constrained	0.477 ± 0.000	0.385 ± 0.000	0.998 ± 0.000	0.488 ± 0.000
p07	Plain ML constrained	0.304 ± 0.061	0.886 ± 0.012	0.956 ± 0.009	0.003 ± 0.002
p07	Plain ML structured constrained	0.440 ± 0.052	0.890 ± 0.012	0.969 ± 0.011	0.007 ± 0.006
p10	Best classical constrained	0.363 ± 0.000	0.322 ± 0.000	0.987 ± 0.000	0.623 ± 0.000
p10	Plain ML constrained	0.526 ± 0.048	0.794 ± 0.158	0.863 ± 0.142	0.072 ± 0.077
p10	Plain ML structured constrained	0.637 ± 0.037	0.770 ± 0.188	0.924 ± 0.065	0.096 ± 0.100

to 0.637 ± 0.037, while staying above the constrained classical baseline for all three seeds. On p07, the structured plain method again improves over plain machine learning on mean support-volume IoU (0.440 ± 0.052 versus 0.304 ± 0.061) and slightly improves mean trace-support overlap and crossline continuity, but the constrained classical baseline retains the highest raw p07 support-volume IoU by occupying a much larger fraction of the allowed support region.

This stability check sharpens the interpretation of the field results. The direct p10 branch supports a robust support-volume claim for structured support mapping. The p07 branch supports a more selective and laterally coherent support-mapping claim relative to plain machine learning, but not a blanket reversal of the constrained classical baseline on raw support-volume IoU.

5.5 Cross-benchmark gate and adaptation-failure analysis

An additional field study was performed to define a deployable operating protocol for the wave-temporal comparator rather than to argue for universal superiority. In low-memory mode with field adaptation steps enabled, adaptation collapsed support occupancy on both branches. For a p10 check at quantile 0.82, support-volume intersection-over-union fell from 0.526 (no adaptation) to 0.000 (adaptation), while the predicted fraction outside support rose from 0.090 to 1.000. For p07 at the same quantile, support-volume intersection-over-union similarly fell from 0.159 to 0.000 and outside-support fraction rose to 1.000. This motivated a fixed-artifact, no-adaptation protocol for the cross-benchmark sweep.

Using that protocol, a three-seed quantile sweep on $q \in [0.74, 0.80]$

Table 5: Cross-benchmark gate for wave-temporal deployment with fixed training artifacts and no field adaptation (three-seed means). Gate condition: p10 support-volume IoU > constrained classical and p07 support-volume IoU ≥ constrained classical.

Quantile	p10 wave SV IoU	p10 classical SV IoU	p07 wave SV IoU	p07 classical SV IoU	p10 margin	p07 margin
0.74	0.570	0.345	0.431	0.407	+0.225	+0.024
0.75	0.570	0.345	0.420	0.407	+0.225	+0.013
0.76	0.570	0.345	0.406	0.407	+0.225	-0.001
0.77	0.570	0.345	0.394	0.407	+0.225	-0.012
0.78	0.570	0.345	0.380	0.407	+0.225	-0.027
0.79	0.570	0.345	0.364	0.407	+0.225	-0.042
0.80	0.570	0.345	0.346	0.407	+0.225	-0.061

was run jointly on p10 and p07. On p10, wave-temporal support-volume intersection-over-union remained above constrained classical for every tested quantile (0.570 versus 0.345). On p07, performance was threshold-sensitive and met or exceeded constrained classical only at quantiles 0.74 and 0.75, with margins of +0.024 and +0.013. Table 5 summarizes the resulting gate decision.

The pass region is therefore narrow but reproducible. Quantile 0.74 is selected as the primary operating point, with 0.75 retained as a sensitivity check. We interpret this as a deployment operating region rather than as a universal threshold rule.

5.6 Negative results and false alarms

Three negative results are important to retain. First, the hybrid model does not outperform plain machine learning on the harder synthetic benchmark and remains weaker on the direct field benchmark. Second, the layer-aware extension does not improve the field result and reduces both trace-support overlap and crossline continuity. Third, low-memory field adaptation can collapse wave-temporal support occupancy in this setting, which makes adaptation safety checks necessary before deployment. These outcomes help define the scope of the contribution supported by the present benchmark.

The negative-control synthetic scenarios also reinforce this point. On mismatch-only and out-of-zone examples, plain machine learning keeps the false-positive rate low while retaining higher Dice than the hybrid model for out-of-zone changes. These scenarios do not replace field validation, but they show that the benchmark can reject weak hypotheses as well as confirm

positive results.

6 Discussion

The main result is that benchmark-constrained structured support reconstruction adds value when the underlying section-level learner is stable, but that value depends on which benchmark question is being asked. On the direct eleven-inline p10 benchmark, the structured method improves support-volume overlap relative to constrained plain machine learning in the main run and retains that ordering across three reused seed artifacts. On the temporal p07 benchmark, the structured method improves on plain machine learning and remains much more selective than the constrained classical baseline, but the classical baseline still attains the highest raw support-volume IoU by occupying too much of the permitted support region. This matters for CCS monitoring because a method that improves section-level overlap while producing spatially inconsistent or overly broad support bodies is unlikely to be useful in practice. In that sense, the paper's novelty lies less in proposing a new learner than in showing how open public products can be assembled into a pseudo-3D benchmark where support mapping and deployment decisions are evaluated explicitly.

A second main result is methodological: explicit operating-point selection materially changes the field conclusion for wave-temporal deployment. With fixed artifacts and no adaptation, p10 remains above constrained classical across the tested quantiles, while p07 crosses the classical baseline only in a narrow region (0.74 and 0.75). This produces a reproducible gate decision rather than a one-off threshold claim and makes the transfer conclusion more defensible for field use.

The comparison with the other tested methods defines the scope of that contribution. The hybrid model does not outperform plain machine learning on the harder synthetic benchmark and remains weaker on the direct field benchmark. The layer-aware extension also fails to improve the field results. In addition, low-memory adaptation can fail catastrophically without explicit safeguards. These outcomes strengthen the study scientifically because they show that the benchmark can reject unsupported method hypotheses as well as identify effective ones. The present evidence therefore supports a benchmark-and-protocol contribution centered on structured support mapping for p10, an honest support-mapping comparison on p07, and a practical

cross-benchmark operating protocol.

Several limitations remain. The field results are benchmark-constrained and should be interpreted as support mapping. The pseudo-3D benchmark is assembled from multiple inlines rather than from the full 3D field cube. The support-volume proxy is benchmark-derived and evaluates support occupancy rather than direct saturation. The synthetic benchmark is more demanding than the earlier version, but it still does not represent full site-scale coupled flow and wave physics. Even with these limitations, the study provides an open-data testbed for comparing field-facing CO_2 monitoring methods under shared constraints and shared metrics.

These findings also have a practical implication for CCS monitoring. Open field evaluation benefits from metrics that reward spatial selectivity and crossline coherence in addition to section-wise agreement. In that setting, structured support reconstruction converts stable section-level probabilities into pseudo-3D support volumes that are easier to assess against public benchmark products and more interpretable as monitoring outputs.

7 Conclusions

This paper introduced an open pseudo-3D benchmark and operating protocol for time-lapse CO_2 support mapping at Sleipner using eleven neighboring inlines, public benchmark constraints, and separate temporal and direct evaluation settings. The benchmark was used to compare constrained classical baselines, plain machine learning, hybrid machine learning, and deterministic support reconstruction stages.

The main findings are straightforward. Plain machine learning is more stable than the hybrid model on the harder three-seed synthetic benchmark. Structured support reconstruction gives the strongest support-volume result on the direct p10 pseudo-3D benchmark and retains that ordering across three reused seed artifacts. On the p07 temporal benchmark, structured support reconstruction improves on plain machine learning and remains far more selective than the constrained classical baseline, but it does not overturn the classical baseline on raw support-volume IoU. The layer-aware extension does not improve the benchmark. A fixed-artifact cross-benchmark gate study further shows that wave-temporal deployment passes both p10 and p07 constrained-classical conditions only at quantiles 0.74 and 0.75, with 0.74 as the primary operating point, and that low-memory adaptation can

collapse the field output if left unchecked. The strongest conclusion supported by the present results is therefore twofold: benchmark-constrained structured support mapping is a strong open p10 baseline and a more selective p07 support-mapping comparator, and explicit operating-point protocols are necessary for reproducible field transfer claims.

The most valuable next step is broader evaluation against additional practical baselines and, if possible, an extension from pseudo-3D support mapping toward more directly quantitative field products.

Data and code availability

The field data used in this study are publicly available through the Sleipner 4D Seismic Dataset and the Sleipner 2019 Benchmark Model on CO2DataShare [8, 9]. The code used to construct the benchmark, run the reported experiments, and generate the manuscript figures and tables is available from the repository associated with this study. The accompanying reproducibility materials include the synthetic seed sweep, the three-seed field-stability tables, and the cross-benchmark operating-point and adaptation-check tables used in the manuscript.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

Generative AI tools were used only to assist with language editing and manuscript organization. All scientific claims, numerical results, interpretations, and final wording were reviewed and verified by the author, who takes full responsibility for the content of the manuscript.

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